Bhavanisankar G S

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.60
Santhanam Vidhyalaya	Class XII (CBSE)	2024	98.40%
Sri Akilandeswari Vidyalaya	Class X (CBSE)	2022	99.00%

SCHOLASTIC ACHIEVEMENTS

Runner-Up, **FIFS Sports Data Gameathon** (*Dream11 & FIFS, 2025; felicitated at Waves Summit 2025*): Represented IIT Hyderabad (Team ML Mavericks) nationally, building an AI/ML model to optimize fantasy cricket lineups and maximize points across the ICC Champions Trophy 2025

A+ grade in **6 Courses**: Awarded to the top 2% of the class; rarely given.

Awarded **Best Outgoing Student** award for Class X and Class XII.

PROJECTS

Machine Learning Framework Design

- Developed a **Feedforward Neural Network** (93% accuracy) and classical classifiers (**LDA, k-NN**) entirely from scratch in **NumPy**, manually deriving backpropagation gradients.
- Implemented a tabular **Q-Learning agent** to solve custom MDPs and applied **Bayesian Linear Regression** to model predictive uncertainty.

AI-Driven Cricket Player Performance Predictor

- Developed a Random Forest regression model in Python to predict future player performance based on historical match data.
- Engineered a custom statistical scoring metric designed to improve prediction accuracy for fantasy team selection.

Subsurface Radar & DSP Pipeline

- Engineered a **time-domain cross-correlation matched filter** from scratch in Python to process simulated radar returns, isolating target reflections buried under **heavy noise (AWGN)**.
- Extracted signal **energy envelopes via Hilbert transform** to compute wave **Time-of-Flight (ToF)**, enabling precise localization of subsurface reflectors.

5-Stage Pipelined RISC-V Processor (RV64IM)

- Designed and implemented a **five-stage pipelined** RISC-V processor core from scratch using Verilog.
- Engineered advanced microarchitectural features, including **data forwarding**, **hazard detection**, and **branch prediction,** to optimize instruction throughput.
- Synthesized the design and deployed the core onto an **Xilinx Spartan-7 FPGA board** for hardware-in-the-loop verification.

Complex Eigenvalue Solver using QR Iteration

- Implemented the classical **QR algorithm** in C using **Gram-Schmidt orthogonalization** to compute eigenvalues of general n×n real and complex matrices.
- Designed a 2×2 **block-extraction routine** to recover complex-conjugate eigenvalue pairs for real matrices with complex eigenvalues.
- Validated results against NumPy's ***eigvals()*** output and profiled runtime to benchmark convergence behavior.

SKILLS

Programming Languages: C | C++ | Python | System Verilog | Verilog | LaTEX | RISCV Assembly

Software & Tools: Linux | Git | GitHub | Xilinx Vivado | Icarus Verilog | GTKWave | Intel Quartus | Vim

RELEVANT COURSES

Mathematics: Probability | Math for EE | Matrix Theory | Vector Calculus | Complex Variables | Differential Equations | Introduction to Statistics

CS & AI: Computer Architecture | Signal Processing | Machine Learning | Data Structures and Algorithms | Introduction To Programming

Electronics: Analog Electronics | Digital Systems | Circuit and Network Analysis | Semiconductor Fundamentals | Electrical Circuits Lab | Electronic Devices and Circuits Lab | PCB Design and Layout for Electronic Systems

POSITIONS OF RESPONSIBILITY

Teaching Assistant | *Circuits and Network Analysis* (EE1101)

- Designed, evaluated, and graded coursework and final assignments.
- Conducted SPICE sessions and provided academic support to resolve student queries.

Digital Systems and Signal Processing Head | *Elektronica IITH*

- Conducted peer-to-peer technical sessions and competed in national-level events like eYRC-Maze Solver 2025 and the IEEESSC Arduino Contest 2025.
- Lead the club, organize technical sessions and competitions, and manage student projects.

Internship Coordinator | *Office of Career Services, IITH*

- Coordinate campus placement drives, verify student profiles, and connect with corporate HRs.
- Pitch to companies for student internships and help design the departmental brochure.

EXTRACURRICULAR

Bronze Medalist, **Carrom** -- *DIESTA* 2025 & 2026

Active member, **National Service Scheme** (**NSS**) -- Participated in Clean India drives, marathons, and community service events.